

FIN305 Risk Management for Business

Week 2 Questions

Practice questions and problems

The numbers in parentheses after the questions correspond to the original numbering used in the textbook, which is intended to assist students in easily finding the relevant questions within the book.

1. Suppose that a bank has \$5 billion of one-year loans and \$20 billion of five-year loans. These are financed by \$15 billion of one-year deposits and \$10 billion of five-year deposits. Explain the impact on the bank's net interest income of interest rates increasing by 1% every year for the next three years. (14.1)
2. What does duration tell you about the sensitivity of a bond portfolio to interest rates? What are the limitations of the duration measure? (14.6)
3. A five-year bond with a yield of 11% (continuously compounded) pays an 8% coupon at the end of each year. (14.7)
 - (a) What is the bond's price?
 - (b) What is the bond's duration?
 - (c) Use the duration to calculate the effect on the bond's price of a 0.2% decrease in its yield.
 - (d) Recalculate the bond's price on the basis of a 10.8% per annum yield and verify that the result is in agreement with your answer to (c).

Solutions

Each week, students should attempt the provided exercise questions after lectures. Solutions can be found at the end of the textbook to check your answers. For those questions to which no solutions can be found in the textbook, the detailed solutions/explanations will be shown in the present file as follows. Selected questions will be discussed in tutorials, where the instructor will offer more detailed solutions. Recordings of the related tutorials will be made available to all students via Learning Mall.